

TUGBA CAGLA EREN

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SUMMARY

Computer Engineering graduate specializing in LLM and RAG applications. Experienced with Python, FastAPI, LangChain, ChromaDB, and Redis for building scalable, production-ready AI services. Active Women in AI Ambassador for Türkiye, passionate about driving GenAI ecosystems.

WORK EXPERIENCE

Basarsoft

Jul 2022 - Sep 2022

Full Stack Developer Intern

Ankara, Turkey

- Developed a full-stack geospatial data management system using .NET Web API, PostgreSQL, OpenLayers, and Node.js. Set up CI/CD pipelines for automated deployment and used Git for version control.
- Improved data synchronization speed by 30% through query optimization. Built monitoring dashboards to track system performance.
- Processed and managed over 500 geospatial data points while ensuring data integrity and system accessibility.

Atosis Arac Takip ve İletişim Sistemleri

Jul 2020 - Aug 2020

Software Engineer Intern

Ankara, Turkey

- Developed a .NET-based data integration system using automated testing workflows to validate data migration reliability, reducing manual data entry validation by 85%.
- Implemented custom parsing logic using .NET Regex and HTTP clients to extract vehicle attributes (price, year, mileage) from semi-structured web pages with 96% accuracy and introduced logging and monitoring to ensure data integrity.

PROJECTS

Genomic-RAG for RARS1 Gene Analysis

Python, LangChain, ChromaDB, Ollama

- Built a local RAG pipeline with LangChain and ChromaDB, implementing automated guardrails and prompt validation techniques to test and ensure reliable genomic data analysis.
- Implemented structured prompting and metadata filtering to ensure answers remain accurate and limited to relevant documents.
- Set up monitoring and observability tools to track retrieval performance, using test data to optimize vector embeddings for higher response accuracy.

AI Fraud Detection System

Python, TensorFlow, Scikit-learn, Pandas, FastAPI, Redis, Streamlit

- Developed a real-time fraud detection service with FastAPI, incorporating automated functional and regression test suites within the CI/CD pipeline for production readiness.
- Designed custom testing scenarios to evaluate machine learning model behavior on highly imbalanced financial datasets, achieving a 0.95 ROC-AUC score.
- Used Redis caching to manage latency, reducing response time by 60% and ensuring fast risk assessments during live transactions.

TECHNICAL SKILLS

- AI & ML:** LLMs (Llama 3, Gpt-4o), RAG, Agentic AI, Prompt Engineering, LangChain, PyTorch, TensorFlow, Scikit-learn
- Backend & APIs:** FastAPI, Rest APIs, WebSockets, .NET
- Programming:** Python, C#, JavaScript
- Databases:** SQL, PostgreSQL, Redis, ChromaDB (Vector DB)
- Tools & Practices:** Git, Docker, CI/CD, Agile, Scrum, Testing

EDUCATION

Kirikkale University

Computer Engineering

Sep 2018 - Feb 2024

Kirikkale

CERTIFICATES

- BTK Academy: Python and TensorFlow for Data Science
- Google Cloud: Introduction to AI & ML Boostcamp
- Data Science & Machine Learning: 100-Day Intensive Camp